

Comparative Study of Traditional and Backpropagation-Free Deep Learning Methods for Anomaly Detection

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Abstract—This project is about anomaly detection on industrial images. We know that anomaly detection is very important for quality control. There are many methods but they couldn't be the most efficient method. Traditional methods like Isolation Forest are fast but not so accurate. Deep learning methods are accurate but their training process is very slow and computationally expensive because of backpropagation. Our hypothesis was that backpropagation free methods like PatchCore and PaDiM can give us both high accuracy and high speed. We tested these methods on MVTec AD datasets carpet, bottle and hazelnut categories. The results showed that our hypothesis is correct. PatchCore and PaDiM achieved almost perfect accuracy (over 98% AUROC) with much less computation time compared to a standard Autoencoder.

Index Terms—anomaly detection, deep learning, PatchCore, PaDiM, MVTec AD, backpropagation free

I. INTRODUCTION

In industrial manufacturing it is very important to find defective products. This is called anomaly detection. For companies, doing this fast and correctly is very important because it saves a lot of money. We can find many computer vision methods for this problems.

In this project we compared different types of anomaly detection methods. We wanted to find a method that is both fast and accurate. This is the main problem because usually fast methods are not accurate, and accurate methods are not fast.

A. Problem Statement

Traditional machine learning methods (for example Isolation Forest or OneClass SVM) can analyze images very quickly. But they usually miss a lot of anomaly, so their accuracy is not enough for real life applications.

However deep learning model like autoencoders can be very accurate. But they need a long training process which can take hours on powerful Gpu. This is because they use backpropagation to learn. This high computational cost can be a big problem for many companies.

So we need a method that has the accuracy of deep learning, but without the high cost of training.

B. Hypothesis

Our hypothesis was simple: we can reach the best of both accuracy and fast. We believed that backpropagation free deep learning methods, specifically PatchCore and PaDiM are the most efficient solution. These methods use pretrained networks to extract features, so they don't need a long training process. We predicted that they could achieve accuracy like heavy deep learning models, but with a much lower computational cost.

II. METHODOLOGY

To test our hypothesis, we selected five methods from three different groups and tested them on three different categories from the MVTec AD dataset. Our main metrics were AUROC for accuracy and Model Preparation time for computational cost.

A. Methods

We grouped our methods as:

- **Traditional Methods:** These methods work on raw pixel data and are computationally cheap
 - *Isolation Forest:* Tries to isolate anomalies by randomly partitioning the data. Anomalies are easier to isolate, so they have shorter paths in the random trees.[1]
 - *One-Class SVM:* Tries to learn a boundary around the normal data. Anything outside this boundary is considered an anomaly.
- **Heavy Deep Learning (Reference):** This method represents a standard deep learning approach that requires training with backpropagation.
 - *Autoencoder:* An encoder-decoder network that is trained to reconstruct normal images. The idea is that it will fail to reconstruct an anomalous image and the high reconstruction error will be our anomaly score.[2]
- **Backpropagation-Free Methods:** These are the methods we test in our hypothesis. They use a pre-trained network to extract features but do not require backpropagation.

- *PatchCore*: It creates a memory bank of normal image patches. For a test image, it finds the distance of its patches to the nearest normal patch in the memory bank.[3]
- *PaDim*: It creates a statistical model for each patch location from normal images. For a test image, it calculates how much its patches deviate from this normal distribution using Mahalanobis distance.[4]

B. Data

We used the MVTec Anomaly Detection (MVTec AD) dataset [1]. This is a standard dataset for benchmarking anomaly detection algorithms for industrial inspection. We selected three categories to see how methods perform on different types of objects and textures:

- **Carpet**: A complex texture, which we expected to be a hard case.
- **Bottle**: A structured object, which we expected to be an easier case.
- **Hazelnut**: A mix of texture and object properties.

All experiment were conducted on Google Colab using an NVIDIA A100 GPU.

III. RESULTS

We ran all five methods on all three categories. The results are summarized in Table I. The table shows the AUROC score for accuracy and the total time it took to prepare the model and run the test (Model Preparation +Test Time)

TABLE I
GENERAL RESULTS TABLE FOR ALL METHODS AND CATEGORIES

Methods	Categories	AUC Scores	Total Time
Isolation Forest	Carpet	0.46	2 min
One Class SVM	Carpet	0.37	4 min
Autoencoder	Carpet	0.31	22 min
PatchCore	Carpet	0.98	10 min
PaDiM	Carpet	1.00	4 min
Isolation Forest	Hazelnut	0.09	2 min
One Class SVM	Hazelnut	0.65	3 min
Autoencoder	Hazelnut	0.89	7 min
PaDiM	Hazelnut	0.99	2 min
PatchCore	Hazelnut	1.00	12 min
Isolation Forest	Bottle	0.69	1 min
One Class SVM	Bottle	0.95	2 min
Autoencoder	Bottle	0.53	10 min
PaDiM	Bottle	1.00	2 min
PatchCore	Bottle	1.00	4 min

To better understand these result we can look at the average performance of method and the average difficulty of categories. The bar charts in fig 1 and fig. 2 visualize these averages

IV. DISCUSSION AND CONCLUSIONS

The results from our experiments strongly support our initial hypothesis.

As we can see from Fig. 1, the backpropagation-free methods, Patchcore and Padim, have a much higher accuracy than

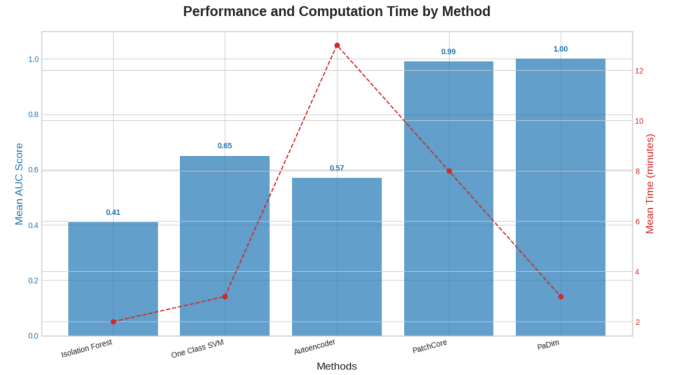


Fig. 1. Performance and Computation Time by Method (Averaged over all categories).

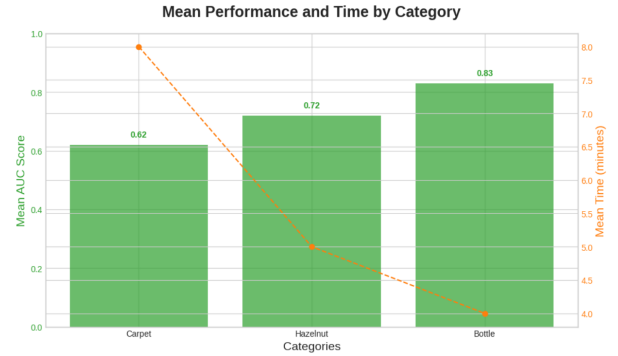


Fig. 2. Mean Performance and Time by Category (Averaged over all methods).

all other methods. Their average AUROC scores are almost perfect (0.99 and 1.00).

The traditional methods, Isolation Forest and One-Class SVM, were very fast but their accuracy was very low. Especially the Isolation Forest results for the ‘hazelnut’ category was interesting. It got an AUROC score of 0.09, which is much worse than a random guess (0.50). This shows that these methods are not reliable for complex image data.

The most interesting result came from our heavy reference model, the AutoEncoder. We expected it to be accurate but slow. However, it was both slow and inaccurate. Its average preparation time was the highest (13 minutes) but its average AUROC score was only 0.57. This makes it the least efficient method in our study. This result show that using a heavy deep learning model with backpropagation is not always a guarantee for high accuracy, especially if the training is not perfect.

In contrast, PatchCore and PaDim are both very accurate and very efficient. PaDim, with a perfect average score and only 3 minutes of average preparation time, seems to be the best overall method in our study This proves our hpothesis: for industrial anomaly detection, backpropagation-free methods that use pretrained networks offer the best balance of high accuracy and low computational cost.

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